JavaScript, the DOM and Asynchronous Code

Web Development - Units 2-3 - Interactivity

Language essentials

- Use const by default and let when reassignment is needed; avoid var.
- Functions are first-class values; arrow functions keep the surrounding this.
- Array methods map, filter and reduce express data transformations clearly.
- Use === instead of == to avoid surprising type coercion.

```
const notes = [{ t: 'ER model', sem: 4 }, { t: 'Sorting', sem: 3 }];
const sem4 = notes.filter(n => n.sem === 4).map(n => n.t);
console.log(sem4);   // ['ER model']
```

The DOM

The Document Object Model is a tree of nodes representing the page. JavaScript selects nodes, changes them and reacts to events.

```
const btn = document.querySelector('#save');
btn.addEventListener('click', () => {
  localStorage.setItem('saved', 'yes');
  btn.textContent = 'Saved';
});
```

Asynchronous JavaScript

JavaScript runs on one thread with an event loop. Slow work such as network requests returns a Promise. async/await makes promise code read top-to-bottom.

```
async function load() {
  try {
    const res = await fetch('/data.json');
    if (!res.ok) throw new Error(res.status);
    return await res.json();
  } catch (err) { console.error(err); }
}
```

Browser storage

- localStorage keeps strings for the same origin until cleared (about 5 MB).
- sessionStorage is cleared when the tab closes.
- Always JSON.parse inside try/catch because stored data can be corrupted.